

Sample Project Documentation

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| Date      | 06 July 2026 | Project Name | Smart Lender       |
| Team Lead | Alla Umesh   | Team Members | DSP, JRJ, LPB, SVB |

Smart Lender Implementation Manual

Project Title: Smart Lender - Credit Risk evaluation platform

Abstract: Financial institutions experience inefficiencies and high turnaround times when auditing applicant loan eligibility. Smart Lender addresses this gap by implementing an automated machine learning classification model built on Flask. The model preprocesses categorical columns using mode imputation and numerical columns using median imputation. It balances target class distributions using SMOTE and applies StandardScaler scaling, achieving high testing accuracy. Real-time evaluation yields predictions along with confidence levels and identifies critical risk factors (low income, poor credit score, high DTI ratios).

Milestone Breakdown Summary

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| Milestone 1 | Model Selection & Architecture: Preprocess and train KNN, Decision Tree, Random Forest, and XGBoost. Best model (XGBoost) selected. |
| Milestone 2 | Core Functionality: Implement single prediction scaling pipelines and risk flagging calculations.                                   |
| Milestone 3 | Flask Backend Development: Define API routes, handle multipart batch CSV streams, and save output locally.                          |
| Milestone 4 | Frontend Development: Build HTML tabs, light-mode CSS, and vanilla JS AJAX request triggers.                                        |
| Milestone 5 | Testing & Verification: Verify scoring accuracy against test_applicants.csv and mock stress tests.                                  |
| Milestone 6 | Project Handover & Documentation: Output comprehensive manuals and Phase PDFs.                                                      |